

Sentinel-1 SAR and **Sentinel-2 Multispectral** are two Earth observation satellite missions operated under the European Space Agency's (ESA) Copernicus program 1. In remote sensing and AI systems like **SatQuery AI**, they represent the two foundational sensor modalities used for land-surface monitoring and cross-modal analysis 1, 2.

1. Sentinel-1 SAR (Synthetic Aperture Radar)

- **Sensor Type:** Active microwave radar (Synthetic Aperture Radar) 2.
- **Key Characteristics:**
 - Transmits its own radar signals and measures the backscatter reflected from Earth's surface 2.
 - **All-Weather & Day/Night Capability:** Penetrates cloud cover, rain, and atmospheric haze, acquiring imagery regardless of sunlight or weather conditions 2.
 - **Structural Data:** Captures physical geometry, surface roughness, structural textures, and moisture/dielectric properties rather than visual colors 2.
 - **Primary Uses:** Disaster monitoring (floods, oil spills), ocean/ice mapping, structural urban change, and surface moisture assessment 2, 3.

2. Sentinel-2 Multispectral (Optical)

- **Sensor Type:** Passive optical multispectral radiometer 2.
- **Key Characteristics:**
 - Captures reflected solar radiation across 13 spectral bands (including visible RGB, Near-Infrared NIR, and Short-Wave Infrared SWIR) 2.
 - **Spectral & Contextual Richness:** Provides detailed color, visual texture, and spectral signatures used to identify land-use/land-cover (LULC) types 2, 4.
 - **Operational Limits:** Requires daylight and clear skies; obscured by clouds, heavy smoke, and darkness 2, 5.
 - **Primary Uses:** Agricultural monitoring, vegetation health tracking, land-cover classification, and environmental analysis 2, 3.

Comparison Summary

Feature, Sentinel-1 SAR, Sentinel-2 Multispectral

Data Modality, Active microwave radar backscatter 2, Passive optical reflected sunlight 2

Primary Information, "Geometry, surface roughness, structural details 2", "Spectral signatures, surface color, visual context 2"

Cloud/Night Penetration, "Yes (penetrates cloud cover, operates day and night) 2", No (blocked by clouds and requires sunlight) 2

Visual Appearance, Greyscale backscatter intensity maps 2, True-color (RGB) and false-color composite images 2

Role in SatQuery AI

In **SatQuery AI**, Sentinel-1 and Sentinel-2 data are used as paired, co-registered observations (primarily sourced from the **BigEarthNet** benchmark dataset) 1, 6, 7:

- **Cross-Modal Fusion:** When optical Sentinel-2 imagery is obscured by cloud cover or shadows, co-registered Sentinel-1 SAR imagery provides the underlying structural

information (such as identifying water bodies or built-up regions through clouds) 2, 4, 5.

- **Complementary Information:** Combining both modalities yields more reliable, grounded conclusions than either sensor can provide individually 2, 8.

💡 *Would you like to see how SatQuery AI's structured fusion pipeline extracts and compares complementary evidence between a Sentinel-1 SAR and Sentinel-2 optical pair?*